OcuDetect: Multi-label Classification of Ocular Diseases using CNNs and Self-Attention

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https://github.com/selinlyx/OcuDetect/tree/main

Introduction

2.2 billion people worldwide are impacted by vision impairment, primarily caused by ocular diseases such as refractive errors, cataracts, diabetic retinopathy, glaucoma, and age-related macular degeneration [1]. Early detection of ocular disease is crucial to preventing irreversible vision loss [2], yet referral misdiagnosis occurs in 49% of ophthalmologic cases, leading to harm in 26% of patients [3]. Additionally, analyzing retinal images from modalities such as colour fundus photography (CFP) and optical coherence tomography (OCT) is time-consuming and requires specialists who are not always accessible [4][5]. These challenges of traditional diagnosis workflows underpin the need for methods that are accurate and efficient. Fortunately, deep learning (DL) models have demonstrated improved diagnostic accuracy and efficiency across multiple medical domains, including ophthalmology. This work aims to improve the accuracy and accessibility of ocular disease diagnosis and combat vision impairment by proposing a convolutional neural network (CNN) based framework called OcuDetect.

Illustration

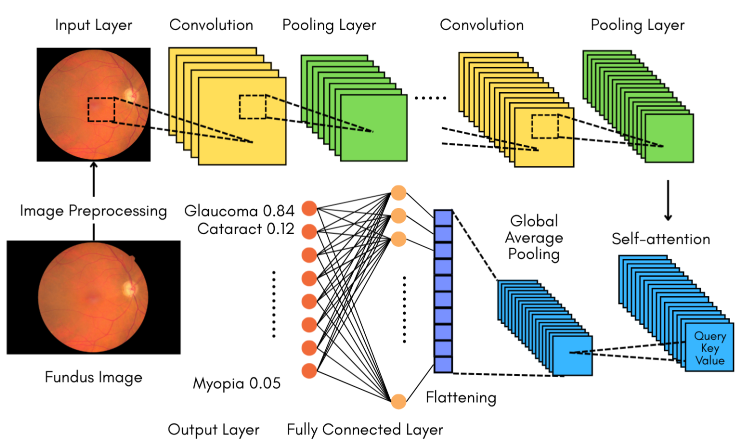

Figure 1: OcuDetect architecture, with a backbone CNN, self-attention, and pooling.

Background & Related Work

Many early DL approaches for ocular disease detection focused on single-label classification with CNNs. For example, Lu et al. (2018) uses a 101-layer ResNet architecture, with transfer learning to detect one of four eye diseases from OCT images with 95.9% accuracy [6]. Later models recognized that eye diseases can be concurrent, and thus the need for multi-label classification in this domain. For instance, DKCNet (2023) augments a backbone CNN with an attention block for discriminative feature attention maps and a squeeze-and-excitation (SE) block to capture inter-channel relationships for multi-label eye disease classification [7].

While CNNs have been the dominant architecture, recent work on vision transformers (ViTs) has demonstrated their ability to leverage self-attention to capture global feature relationships [8]. Given this advantage, some ViTs outperform CNNs in ocular disease classification, such as Cutur and Inan's model achieving a 98.1% accuracy [8].

Despite the progress, broad-spectrum classification can remain a challenge. Rieck et al. achieved an accuracy of 82.52% classifying from 9 ocular diseases using a ResNet-50 architecture with transfer learning and Gaussian filtering [4]. Additionally, many models still face challenges generalizing across patient populations, disease stages, and imaging modalities [9].

Data Processing

Many CFP ocular disease datasets are available in the JPEG format, including ODIR [10] (8 classes, 8000 images), Eye Disease Image Dataset (9 classes, 5335 images) [11], and BRSET (13 classes, 16 266 images) [12]. The following preprocessing steps will be taken:

- **Cropping to the region of interest:** CFPs have a circular field of view; the background has no diagnostic value and can lead to false positives [13]. Thus, background pixels will be blackened using the Fundus Circle Cropping tool by Berens Lab [14].
- **Image rescaling:** images shall be uniform in pixel size to comply with the fixed input size.
- **Contrast enhancement:** CFPs can have low contrast, obscuring important diagnostic features like blood vessels [15]. Thus, the Contrast Limited Adaptive Histogram Equalization tool by OpenCV will be used to improve contrast.
- **Normalization:** will be employed to achieve uniform pixel intensity values across all images, ranging from 0 to 1.
- **Data splitting:** the pipeline will use a 70% train, 15% validation, and 15% test split.

Architecture

The proposed architecture (Figure 1) of OcuDetect employs the pretrained EfficientNet-B0 as a backbone with multiple convolutional layers. Due to the resource and time constraints, EfficientNet-B0 is selected for its lightweight architecture and strong performance-to-parameter ratio [16]. Self-attention can benefit multi-label classification, as some ocular diseases are characterized by their non-local dependencies. For example, the relative spatial arrangement of patterns like microaneurysms and exudates can indicate the diabetic retinopathy class [17]. ViTs require large datasets to generalize [18], making them ill-suited for the small CFP datasets available; thus, a self-attention block is added to the CNN backbone. A Global Average Pooling layer is applied after the self-attention block, followed by fully connected layers with ReLU activation. The output layer uses a sigmoid activation function for multi-label classification. Binary cross-entropy loss and Adam optimizer will be used.

Baseline Model

ResNet-50 is a common backbone in ocular disease classification research, often achieving high accuracy [19]. A ResNet-50 with the default ImageNet classifier head and no self-attention block will be used as a baseline to evaluate the effects of adding self-attention and a custom classifier head. The same datasets will be used to train and test all models for comparison.

Ethical Considerations

Using DL for ocular disease diagnosis carries many ethical considerations. Algorithmic bias arises from a lack of ethnic diversity in the CFP training datasets, which neglects the variability in retinal features across different ethnicities. This means the prediction results may generalize poorly to underrepresented ethnic groups. Furthermore, the retinal vasculature is a unique biometric identifier, making data use in training DL models a potential privacy risk [20]. Finally, over-reliance on such models can lead to catastrophic misdiagnosis that causes patient harm in preventing the appropriate care.

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